Events and Listeners in Servlet Programming

Kishan

Q) What is an Event?

Ans) An event represents an action detected by a program. In servlets, events are generated when certain actions occur on servlet objects like **S**ervletContext, **S**ervletRequest, and **H**ttpSession

Q) What is a Listener?

Ans) A listener is an object that listens for events and processes them when they occur. In servlet programming, listeners are often used to monitor specific actions (like the creation or destruction of sessions or requests) and execute certain operations in response to these events.

Servlet listeners can be registered to monitor events on the following objects:

- ServletContext - Monitors events in the servlet's environment.
- ServletRequest - Monitors request-level events.
- HttpSession - Monitors session-level events.

1) ServletContext Listeners

ServletContextListener : Monitors the lifecycle of the ServletContext

Methods:

contextInitialized(ServletContextEvent): Called when the ServletContext is initialized.

contextDestroyed(ServletContextEvent): Called when the ServletContext is destroyed.

Kishan

ServletContextAttributeListener : Monitors changes to attributes in the ServletContext

Methods:

attributeAdded(ServletContextAttributeEvent): Called when an attribute is added.

attributeRemoved(ServletContextAttributeEvent): Called when an attribute is removed.

attributeReplaced(ServletContextAttributeEvent): Called when an attribute is replaced.

2) ServletRequest Listeners

ServletRequestListener : Monitors the lifecycle of ServletRequest

Methods:

requestInitialized(ServletRequestEvent): Called when a request is created.

requestDestroyed(ServletRequestEvent): Called when a request is destroyed.

Kishan

ServletRequestAttributeListener : Monitors changes to attributes in the ServletRequest

Methods:

attributeAdded(ServletRequestAttributeEvent): Called when an attribute is added.

attributeRemoved(ServletRequestAttributeEvent): Called when an attribute is removed.

attributeReplaced(ServletRequestAttributeEvent): Called when an attribute is replaced.

3) HttpSession Listeners

HttpSessionListener : Monitors the lifecycle of HttpSession

Methods:

sessionCreated(HttpSessionEvent): Called when a session is created.
sessionDestroyed(HttpSessionEvent): Called when a session is destroyed.

HttpSessionAttributeListener : Monitors changes to attributes in the HttpSession

Methods:

attributeAdded(HttpSessionBindingEvent): Called when an attribute is added.
attributeRemoved(HttpSessionBindingEvent): Called when an attribute is removed.
attributeReplaced(HttpSessionBindingEvent): Called when an attribute is replaced.

Kishan

Annotations in Java

Kishan

Annotations in Java are metadata tags that can be added to Java programming components such as variables, methods, classes, and interfaces. **They provide additional information to the compiler or the JVM**, which helps during compilation or runtime. Annotations in Java start with the @ symbol.

@Override :	Indicates that a method is intended to override a method in a superclass
@SuppressWarnings :	Often used to ignore warnings about unchecked operations or deprecated code. Improves code readability by removing unnecessary warnings
@FunctionalInterface :	Marks an interface as a functional interface, which should contain exactly one abstract method.
@WebServlet :	Used in servlets to define the URL pattern associated with a servlet
@WebFilter :	Used to define URL patterns for which the filter should be executed.
@WebListener :	Identifies a listener in Java EE applications, such as a session, request or context listener

Kishan

Servlet Life Cycle

Kishen

The life cycle of a servlet represents the various stages a servlet goes through from the start of its life to its termination. The servlet life cycle is managed by the servlet container and includes these following stages:

1) ***Loading:*** Locates and loads the servlet class based on the URL pattern.
2) ***Instantiation:*** Creates an instance of the servlet class.
3) ***Initialization:*** Calls init() for setup, executed only once.
4) ***Request Handling:*** Handles client requests through service() or HTTP-specific methods.
5) ***Destroying:*** Calls destroy() to clean up resources before the servlet is removed from memory.

Kishan

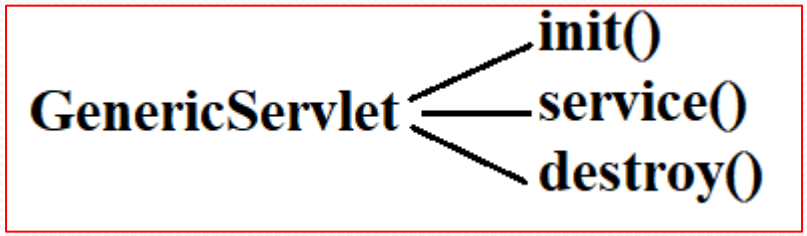
Loading Process:

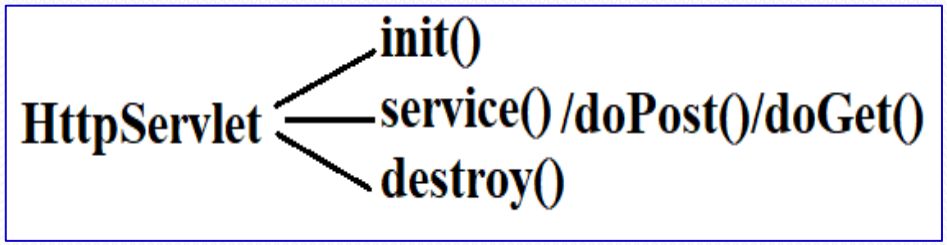
This is the first stage, where the servlet container locates the servlet class using its URL pattern. Once located, the servlet class is loaded into memory, preparing it for execution.

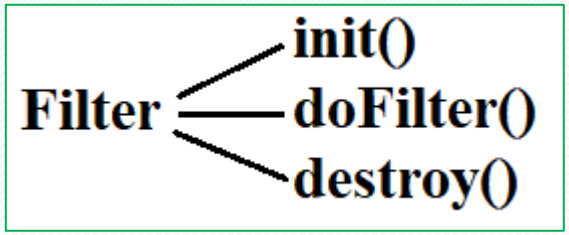
Instantiation Process:

After loading, the servlet container creates an instance of the servlet class. This process is called instantiation. After instantiation, the container recognizes the lifecycle methods that need to be invoked for the respective servlet.

Kishan







Initialization Process:

During initialization, the servlet container calls the init() method to prepare the servlet for handling requests through the service() method. **The init() method runs only once in the servlet's life cycle**, ensuring initialization is performed just once.

Request Handling Process:

This stage involves accepting requests from clients and sending respective responses. The **service()** method or specific HTTP methods like **doGet()** or **doPost()** are used to handle the requests. For every request, the service() method is called, which can handle multiple requests simultaneously from multiple users.

Kishan

Destroying Process:

This final stage involves cleaning up resources used by the servlet during request handling, such as closing database connections, files, or other resources. The servlet container calls the destroy() method **only once** after all requests have been served, and the servlet is about to be removed from memory.

Kishan